

SRIRAM AKELLA

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Webpage: <https://sriramakella7.github.io>

EDUCATION

Tata Institute of Fundamental Research Integrated M.Sc.-Ph.D. in Physics	Aug 2021 – Present
Chennai Mathematical Institute B.Sc. (Hons.) in Mathematics & Physics	Aug 2018 – Apr 2021

RESEARCH INTERESTS

Quantum Information Theory, Quantum Field Theory, and Quantum Gravity.

PREPRINTS AND PUBLICATIONS

1. Akella, S., Thapliyal, K., Mani, H.S. and Pathak, A., 2022. Dynamics of single-mode nonclassicalities and quantum correlations in the Jaynes–Cummings model. [JOSA B](#), **39**(7), pp.1829-1838. [arXiv:2203.04119](#) [quant-ph].

RELEVANT COURSEWORK

Quantum Field Theory II <i>Grade: A+</i>	Aug 2023 – Dec 2023 <i>Tata Institute of Fundamental Research</i>
Quantum Field Theory I <i>Grade: A+</i>	Jan 2023 – Apr 2023 <i>International Centre for Theoretical Sciences</i>
Black Hole Information Paradox <i>Grade: A</i>	Jan 2023 – Apr 2023 <i>Tata Institute of Fundamental Research</i>
Effective Theories and Effective Field Theories <i>Grade: B+</i>	Aug 2022 – Nov 2022 <i>Tata Institute of Fundamental Research</i>
Quantum Information Theory <i>Grade: 10/10</i>	Jan 2021 – Apr 2021 <i>Institute of Mathematical Sciences</i>
Quantum Field Theory <i>Grade: 9/10</i>	Jan 2021 – Apr 2021 <i>Chennai Mathematical Institute</i>
Bell's Theorem and Entanglement <i>Grade: 10/10</i>	Aug 2020 – Nov 2020 <i>Chennai Mathematical Institute</i>

RESEARCH EXPERIENCE

Multipartite measures in chaotic many-body systems <i>with Prof. Abhijit Gadde</i>	Oct 2024 – Present
Exploring the possibility of extending the entanglement membrane picture that appears in chaotic many-body systems to the context of multipartite entanglement measures. Might help in understanding how such systems approach thermal equilibrium.	
Multipartite entanglement measures in stabilizer states <i>with Prof. Abhijit Gadde</i>	Mar 2023 – Present

Devised an efficient algorithm to numerically compute multipartite entanglement measures in stabilizer states.

Multi-entropy in Chern-Simons Theory
with Prof. Abhijit Gadde

Aug 2022 – Present

Calculated multi-entropy of some simple spatial regions in Chern-Simons theory using surgery. Work in progress to develop tools to handle non-trivial spatial regions.

Dynamics of Nonclassicality in the Jaynes-Cummings Model Jan 2021 – Jul 2021
with Prof. Anirban Pathak

Showed that the atom and the photon field exchange nonclassicality through the atom-field correlations under Jaynes-Cummings dynamics. Work in progress to extend these results to noisy Jaynes-Cummings models.

TALKS AND PRESENTATIONS

Multi-entropy in graph states July 2023

Presented our algorithm to numerically compute multi-entropy in graph states as part of my departmental project. Presented examples checking the conjectured monotonicity properties.

The Entropy of Hawking Radiation Apr 2023

Presented Almheiri et. al.'s paper titled "Replica Wormholes and the Entropy of Hawking Radiation" as part of a reading course on the Black Hole Information Paradox.

Chemistry of Superheavy elements Mar 2023

Gave a short talk based on a paper by Schädel titled "Chemistry of the Superheavy elements" as part of a course on Nuclear Physics.

Multi-entropy in Chern-Simons theory Jan 2023

Presented our calculations of multi-entropy in Chern-Simons theory as part of my departmental project.

Phonons as Goldstone Bosons Dec 2022

Presented H. Leutwyler's paper titled "Phonons as Goldstone bosons" as part of a course on Effective Field Theories.

Anyons and Chern-Simons Theory Dec 2022

Gave a short talk on Anyons and Chern-Simons Theory as part of a course on Advanced Quantum Mechanics.

Landauer's Erasure and Exorcism Apr 2022

Gave a short talk on Petrosyan et. al.'s paper titled "Experimental verification of Landauer's principle linking information and thermodynamics" as part of a course on Statistical Physics.

Noether's Second Theorem Mar 2022

Gave a short talk on Noether's second theorem as part of a course on Electrodynamics.

Grover's Algorithm: An Optical Approach Apr 2021

Presented P. G. Kwiat's experimental realization of Grover's algorithm using photons as part of a course on Quantum Information Theory.

Particle Creation in Expanding Universes

Mar 2021

Presented Parker's classic paper titled "Quantized Fields and Particle Creation in Expanding Universes. I" as part of a course on Quantum Field Theory.

Multiplicative Bell Inequalities

Nov 2020

Presented Eliahu Cohen et.al.'s paper titled "Multiplicative Bell Inequalities" as part of a course on Bell's Theorem and Entanglement.

Quantum Entanglement in Time

Oct 2020

Presented Vlatko Vedral et. al.'s paper titled "Quantum Entanglement in Time" as part of a course on Bell's Theorem and Entanglement.

The Thermodynamic Theory of Black Holes

Oct 2020

Presented P. C. W. Davies's classic paper titled "The Thermodynamic theory of black holes" as part of a course on Statistical Physics.

CONFERENCES

Quantum Information, Quantum Field Theory and Gravity

August 2024

Offline

ICTS, Bengaluru

Indian Strings Meeting 2023

Dec 2023

Offline

IIT Bombay

It from Qubit 2023

July 2023

Online

Perimeter Institute for Theoretical Physics

Strings 2023

July 2023

Online

Perimeter Institute for Theoretical Physics

TEACHING

Mathematical Methods

August 2022 – December 2022

Tutor

Tata Institute of Fundamental Research

Classical Mechanics

August 2020 – December 2020

Tutor

Chennai Mathematical Institute

REFERENCES

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